

CS449 / CS549 Midterm 2

I. QUESTION

Given the following grammar L_1 below:

$S \rightarrow NP VP$
 $NP \rightarrow Pronoun$
 $NP \rightarrow ProperNoun$
 $NP \rightarrow Det Nominal$
 $Nominal \rightarrow Nominal Noun$
 $Nominal \rightarrow Noun$
 $VP \rightarrow Verb$
 $VP \rightarrow Verb NP$
 $VP \rightarrow Verb NP PP$
 $VP \rightarrow Verb PP$
 $VP \rightarrow VP PP$
 $PP \rightarrow Prep NP$
 $Noun \rightarrow \dots$
 $Verb \rightarrow \dots$
 $Pronoun \rightarrow \dots$
 $ProperNoun \rightarrow \dots$
 $Det \rightarrow \dots$
 $Prep \rightarrow \dots$

- Fill the spaces with appropriate entries to create a grammar for which a parse tree for "Alaska is a morning trip" can be constructed.
- Construct the parse tree for that parse tree in a.

II. QUESTION

Given the following CFG L_2 in CNF form:

$S \rightarrow NP VP$
 $NP \rightarrow Det Nominal$
 $Nominal \rightarrow Noun Noun$
 $VP \rightarrow Verb NP$
 $VP \rightarrow Verb PP$
 $VP \rightarrow VP PP$
 $PP \rightarrow Prep NP$
 $Noun \rightarrow flight, trip, super, morning$

$Verb \rightarrow is$

$Det \rightarrow the, this$

$Prep \rightarrow on$

Parse the sentence "The morning flight is on this super trip" using CKY parsing. Show your work.

III. QUESTION

Given the following PCFG L_3 in CNF form:

$S \rightarrow NP VP[1]$
 $NP \rightarrow Det Nominal[0.6]$
 $NP \rightarrow Det Noun[0.4]$
 $Nominal \rightarrow Noun Noun[0.8]$
 $Nominal \rightarrow Nominal Noun[0.2]$
 $VP \rightarrow Verb NP[0.5]$
 $VP \rightarrow Verb PP[0.3]$
 $VP \rightarrow VP PP[0.2]$
 $PP \rightarrow Prep NP[1]$
 $Noun \rightarrow flight[0.5]$
 $Noun \rightarrow trip[0.2]$
 $Noun \rightarrow super[0.2]$
 $Noun \rightarrow morning[0.1]$
 $Verb \rightarrow is[1]$
 $Det \rightarrow the[0.8]$
 $Det \rightarrow this[0.2]$
 $Prep \rightarrow on[1]$

Parse the sentence "The morning flight is on this super trip" using probabilistic CKY parsing. Show your work.

IV. QUESTION

- Draw the computation graph for the function $L(w, x, b, u, c) = \text{sigmoid}(w*x + b)*u + c$.
- Calculate the derivatives with respect to every variable in the graph for $w = 1$, $x = 2$, $b = -2$, $u = 4$, $c = -1$.

V. QUESTION

Draw a single input, single hidden layer (with a single hidden node) neural network (multilayer perceptron) for two-class classification problem. Derive the update equation for

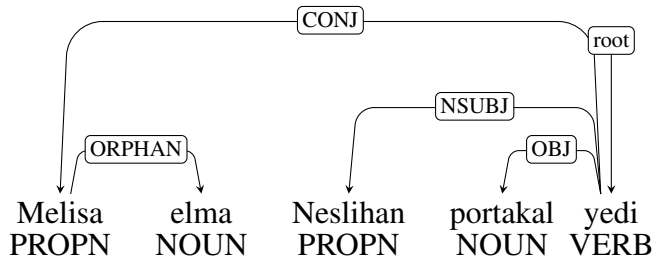
- the weight between the input unit and hidden layer
- the weight between the hidden unit and the output.

Do not forget the bias unit.

VI. QUESTION

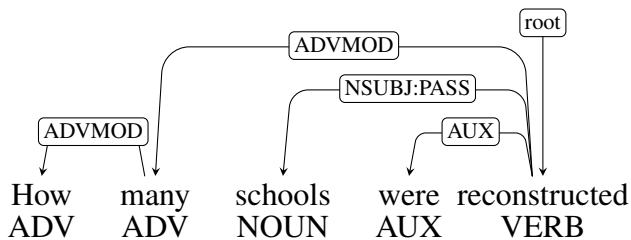
Draw the LSTM network for the sentiment analysis problem with three classes.

VII. QUESTION



Construct the trace of the given transition-based parse above. Show stack, word list, action and relation added at each iteration. Assume that arc-standard parsing has been used.

VIII. QUESTION



Construct the trace of the given transition-based parse above. Show stack, word list, action and relation added at each iteration. Assume that arc-eager parsing has been used.